Shyam Sivan
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EDUCATION

Georgia Institute of Technology Atlanta, GA
B.S. in Aerospace Engineering, *Minor in Astrophysics | GPA: 3.43* *May 2028*
Relevant Coursework: Statics, Dynamics, Mechanics of Materials, Aerodynamics, Thermodynamics, Fluid Mechanics, Aerospace Experimental Methods, Engineering Materials, Engineering Graphics, Differential Equations, Circuits and Electronics, Classical Mechanics

EXPERIENCE

Propulsive Landers at Georgia Tech Atlanta, GA
VTVL Hopper Propulsion Member January 2026 - Present

- Collaborated on developing propulsion system for a team aiming to become the first organization to make an untethered VTVL hybrid rocket aiming to achieve a 50m hop, supporting development of 1000N engine and flight hardware operating at pressures above 300 bar
- Created propulsion P&IDs documenting valves, regulators, pressure transducers, relief devices, instrumentation, and feed system architecture to certify physical plumbing and fabrication aligned with ideal system connections
- Assembled tubing and verified valve configurations with propulsion P&IDs to ensure hardware integration and test readiness
- Developed CAD models for ground test and finalized lander body dimension and configurations, applying GD&T and coordinating interfaces across 4 subteams (propulsion, structures, avionics, GNC) to define clearance and mounting locations before fabrication
- Analyzed post hot fire data from 3 tests, identifying anomalies with the combustion chamber and notifying the combustion chamber and nozzle teams on hardware adjustments allowing efficient residue purging needed for proper combustion flow
- Compared 4 injector geometries through literature, SolidWorks, and ANSYS Fluent, evaluating atomization behavior, pressure loss, manufacturability, and combustion characteristic differences, selecting a coaxial injector based on manufacturability and flow constraints

Smartsoft International Inc. Johns Creek, GA
Digital Transformation Analyst Intern June 2024 – August 2024

- Developed a customer sentiment analysis prototype at an IT consulting firm specializing in SAP business solutions, contributing to a BERT machine learning model achieving around 90% classification accuracy to help clients interpret customer feedback
- Built dashboard components displaying sentiment distributions, common keywords, and downloadable reports, enabling businesses to examine customer feedback by category, date, and location and identify recurring customer concerns and sentiment trends
- Demonstrated prototype to Detroit Power, using live customer feedback to highlight when and where customers reported power issues

RESEARCH

Daedalus Lab at Georgia Tech Atlanta, GA
Injection Molding and Mechanical Recycling Researcher August 2026 – Present

- Researched repeated mechanical recycling cycles' effects on recycled PET and LDPE long-term tensile strength and stress-strain data
- Operated injection molding equipment to produce dog bone specimens for mechanical testing of recycled PET/LDPE material reliability
- Prepared recycled PET and LDPE specimens for characterization through density, viscosity, melt-flow rate, stiffness, and toughness measurements to evaluate how recycling affects mechanical performance, assessing material suitability for repeated reuse

PET Voxel Development Researcher May 2026 – August 2026

- Investigated the mechanical properties of recycled PET-based 3D printing filament through voxel-based specimen design to provide a sustainable and lightweight alternative to current filaments for off-planet structural applications
- Reduced material waste by approximately 20% by optimizing our material extraction device and 3D printer settings through iterative testing across multiple plastic feedstocks and material use-cases, increasing the proportion of feedstock converted to viable material
- Designed CAD fixtures to standardize extraction from recycled plastic pieces and improve structural stability of our extraction device, increasing sample consistency and enabling repeatable structural testing across many experimental batches
- Created dimensioned engineering drawings for mechanical components, documenting geometry, tolerances, and assembly requirements to allow for replicability of piece creation and further iteration of experimental design
- Ran compression tests across 3 materials (PLA, PETG, PET) and 3 scales (1x1 to 3x3), with load capacities of up to 1.4 kN for PLA and 1.2kN for our PET prototype, verifying comparable material strengths and appropriate alternative usage of recycled PET for PETG

Computational Combustion Lab at Georgia Tech Atlanta, GA
Combustion Engine CFD Researcher January 2026 – February 2026

- Compared temperature and velocity profiles of simulated engines with reference data, quantifying agreement with published literature to assess model predictions under conditions such as different operating pressures, inlet velocities, and inlet temperatures
- Developed Python scripts to extract temperature and axial velocities from 10 simulation outputs and generate consistent comparison plots, reducing manual post-processing time from 20 to 7 minutes
- Analyzed 3 simulation cases in ParaView to characterize how inlet velocity affected axial velocity and temperature distribution

SKILLS

Software: SolidWorks (CSWA Certification), GMAT, ANSYS Fluent, ANSYS Mechanical, OpenRocket, MATLAB, Python, Java, Git, ParaView, LabVIEW, Microsoft Office
Design: GD&T, DFM, P&ID Development, Technical Drawings, Soldering, Assembly Modeling, Experimental Design, Machine Shop Tools, 3D Printing (Bambu, Prusa)
Technical: Hybrid Rocket Systems, Feed Systems, Injector Design, Propulsion Systems, Structural/Failure Analysis, Combustion & Fluid Dynamics, Machine Learning